

JOÃO GASPAR

CONTACT

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Aveiro, Portugal

LANGUAGES

Portuguese (native)
English (fluent)
Spanish (intermediate)
French (basic)

PROFILE

MSc candidate in Computer Engineering and Telematics at the University of Aveiro, currently developing a thesis on astronomical data compression. Designed and implemented a C++ compression model for raw telescope data, using arithmetic coding and range coding, with Bash scripts on Linux for experimentation and Python for data visualisation and image conversion. Experienced in performance analysis, scientific data processing and web development, with a strong interest in automation and reliable software systems.

RESEARCH EXPERIENCE

MSc Thesis - Astronomical Data Compression **2025 - PRESENT**
University of Aveiro *AVEIRO, PORTUGAL*

- Designed and implemented a C++ compression model for raw telescope data, integrating arithmetic coding and range coding.
- Evaluated datasets from recognised telescopes, including the Isaac Newton Telescope (INT), measuring bitrate, compression ratio, execution time and memory usage.
- Used Bash/Linux to run experiments and Python to generate plots and convert images from RAW to PGM and FITS.
- Achieved better compression performance than the existing compression tools evaluated in the thesis.

EXPERIENCE

IT and Communications **2024 - PRESENT**
AETTUA - University of Aveiro *AVEIRO, PORTUGAL*

- Contribute to the maintenance and improvement of the association's React-based website.
- Support digital publication of academic projects and events, collaborating on content, planning and technical website tasks.

Freelance Web Developer & Academic Data Projects **2020 - 2024**
Independent *AVEIRO, PORTUGAL*

- Developed websites for independent projects and users.
- Completed academic data-analysis projects using Python; debugged software and documented technical work.

TECHNICAL SKILLS

Programming – C++, Python, Bash/Shell, Java, JavaScript, MATLAB
Web – React, Django, HTML, CSS, REST APIs
Data – Data visualisation, compression benchmarking
Tools and systems – Linux, Git, GitHub, Docker, Jira, Azure DevOps, debugging, technical documentation
Databases – PostgreSQL, MySQL
Concepts – High-performance computing, cloud computing, embedded systems

EDUCATION

MSc in Computer Engineering and Telematics **2024 - EXPECTED 2026**
University of Aveiro *AVEIRO, PORTUGAL*

BSc in Computer and Informatics Engineering **2021 - 2024**
University of Aveiro *AVEIRO, PORTUGAL*